

E1 Basic trigonometry

Name: _____

Class: _____

Date: _____

Time: **278 minutes**

Marks: **231 marks**

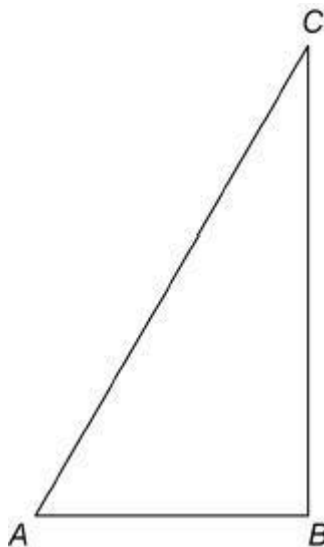
Comments:

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Q1.

ABC is a right-angled triangle.



D is the point on hypotenuse AC such that $AD = AB$.

The area of $\triangle ABD$ is equal to half that of $\triangle ABC$.

(a) Show that $\tan A = 2 \sin A$ (4)

(b) (i) Show that the equation given in part (a) has two solutions for $0^\circ \leq A \leq 90^\circ$ (2)

(ii) State the solution which is appropriate in this context. (1)

(Total 7 marks)

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Q2.

Solve the equation $\tan^2 2\theta - 3 = 0$ giving all the solutions for $0^\circ \leq \theta \leq 360^\circ$

(Total 4 marks)

Q3.

A periodic sequence is defined by $U_n = \sin\left(\frac{n\pi}{2}\right)$

State the period of this sequence.

Circle your answer.

8
 2π
4
 π

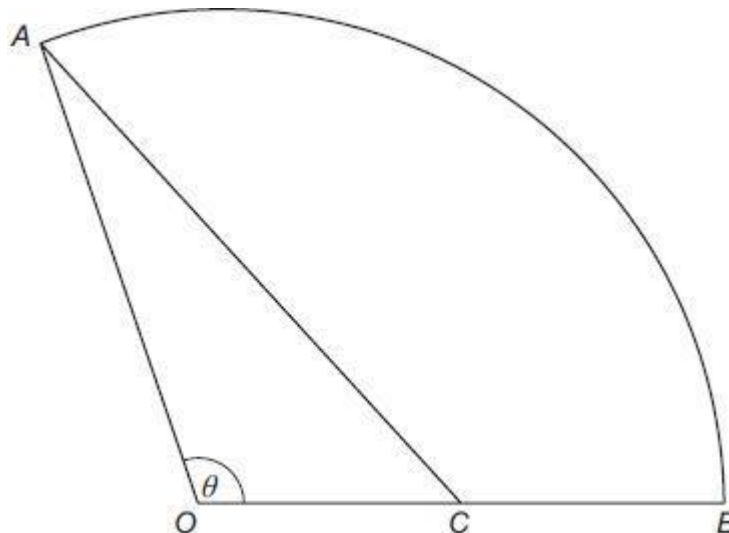
(Total 1 mark)

Q4.

The diagram shows a sector of a circle OAB .

C is the midpoint of OB .

Angle AOB is θ radians.



- (a) Given that the area of the triangle OAC is equal to one quarter of the area of the sector OAB , show that $\theta = 2 \sin \theta$

(4)

- (b) Use the Newton-Raphson method with $\theta_1 = \pi$, to find θ_3 as an approximation for θ . Give your answer correct to five decimal places.

(3)

- (c) Given that $\theta = 1.89549$ to five decimal places, find an estimate for the percentage error in the approximation found in part (b).

(1)

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(Total 8 marks)

Q5.

A parallelogram has sides of length 6 cm and 4.5 cm.

The larger interior angles of the parallelogram have size α

Given that the area of the parallelogram is 24 cm^2 , find the exact value of $\tan \alpha$

(Total 4 marks)

Q6.

Solve the equation

$$\sin \theta \tan \theta + 2 \sin \theta = 3 \cos \theta \quad \text{where } \cos \theta \neq 0$$

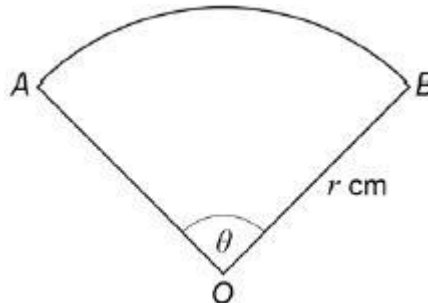
Give **all** values of θ to the nearest degree in the interval $0^\circ < \theta < 180^\circ$

Fully justify your answer.

(Total 5 marks)

Q7.

The diagram shows a sector AOB of a circle with centre O and radius r cm.



The angle AOB is θ radians.

The sector has area 9 cm^2 and perimeter 15 cm .

(a) Show that r satisfies the equation $2r^2 - 15r + 18 = 0$

(4)

(b) Find the value of θ . Explain why it is the only possible value.

(4)

(Total 8 marks)

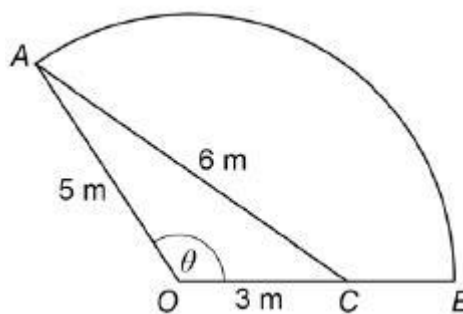
Q8.

Prove the identity $\cot^2 \theta - \cos^2 \theta \equiv \cot^2 \theta \cos^2 \theta$

(Total 3 marks)

Q9.

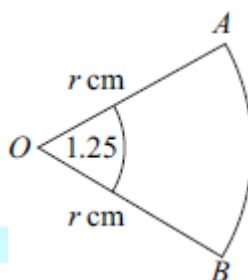
A wooden frame is to be made to support some garden decking. The frame is to be in the shape of a sector of a circle. The sector OAB is shown in the diagram, with a wooden plank AC added to the frame for strength. OA makes an angle of θ with OB .



- (a) Show that the exact value of $\sin\theta$ is $\frac{4\sqrt{14}}{15}$ (3)
- (b) Write down the value of θ in radians to 3 significant figures. (1)
- (c) Find the area of the garden that will be covered by the decking. (2)
- (Total 6 marks)

Q10.

The diagram shows a sector OAB of a circle with centre O and radius r cm.

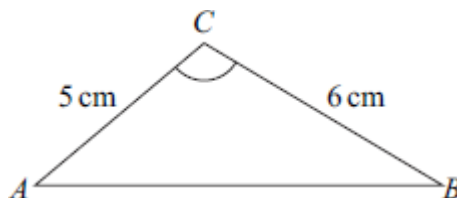


The angle AOB is 1.25 radians. The perimeter of the sector is 39 cm.

- (a) Show that $r = 12$. (3)
- (b) Calculate the area of the sector OAB . (2)
- (Total 5 marks)

Q11.

The diagram shows a triangle ABC .



The lengths of AC and BC are 5 cm and 6 cm respectively.

The area of triangle ABC is 12.5 cm^2 , and angle ACB is **obtuse**.

- (a) Find the size of angle ACB , giving your answer to the nearest 0.1° . (3)

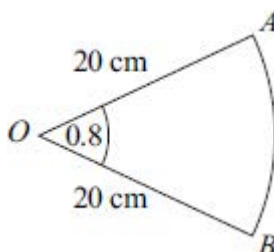
- (b) Find the length of AB , giving your answer to two significant figures.

(3)

(Total 6 marks)

Q12.

The diagram shows a sector OAB of a circle with centre O .



The radius of the circle is 20 cm and the angle $AOB = 0.8$ radians.

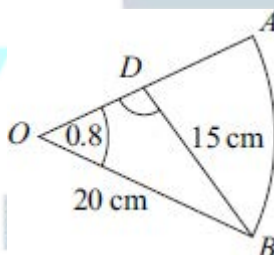
- (a) Find the length of the arc AB .

(2)

- (b) Find the area of the sector OAB .

(2)

- (c) A line from B meets the radius OA at the point D , as shown in the diagram below.



The length of BD is 15 cm. Find the size of the **obtuse** angle ODB , in **radians**, giving your answer to three significant figures.

(4)

(Total 8 marks)

Q13.

By forming and solving a quadratic equation, solve the equation

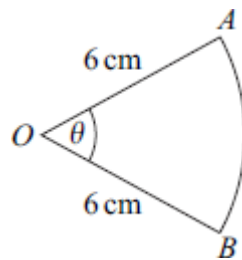
$$8 \sec x - 2 \sec^2 x = \tan^2 x - 2$$

in the interval $0 < x < 2\pi$, giving the values of x in radians to three significant figures.

(Total 7 marks)

Q14.

The diagram shows a sector OAB of a circle with centre O and radius 6 cm.



The angle between the radii OA and OB is θ radians.

The area of the sector OAB is 21.6 cm^2 .

(a) Find the value of θ .

(2)

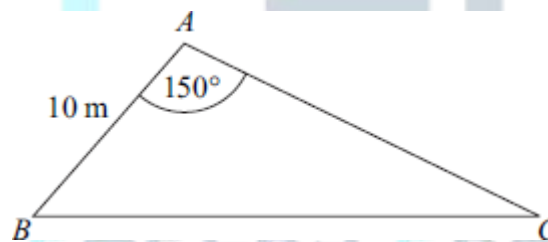
(b) Find the length of the arc AB .

(2)

(Total 4 marks)

Q15.

The triangle ABC , shown in the diagram, is such that AB is 10 metres and angle BAC is 150° .



The area of triangle ABC is 40 m^2 .

(a) Show that the length of AC is 16 metres.

(2)

(b) Calculate the length of BC , giving your answer, in metres, to two decimal places.

(3)

(a) Calculate the smallest angle of triangle ABC , giving your answer to the nearest 0.1° .

(3)

(Total 8 marks)

Q16.

Angle α is acute and $\cos \alpha = \frac{3}{5}$. Angle β is **obtuse** and $\sin \beta = \frac{1}{2}$.

- (a) (i) Find the value of $\tan \alpha$ as a fraction.

(1)

- (ii) Find the value of $\tan \beta$ in surd form.

(2)

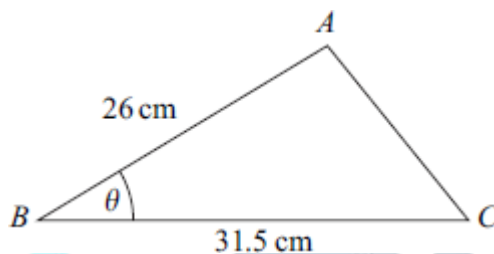
- (b) Hence show that $\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$, where m and n are integers.

(3)

(Total 6 marks)

Q17.

The triangle ABC , shown in the diagram, is such that $AB = 26$ cm and $BC = 31.5$ cm.



The acute angle ABC is θ , where $\sin \theta = \frac{5}{13}$.

- (a) Calculate the area of triangle ABC .

(2)

- (b) Find the exact value of $\cos \theta$.

(1)

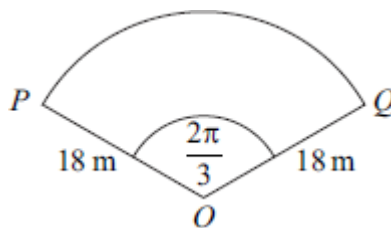
- (c) Calculate the length of AC .

(3)

(Total 6 marks)

Q18.

The diagram shows a sector OPQ of a circle with centre O .

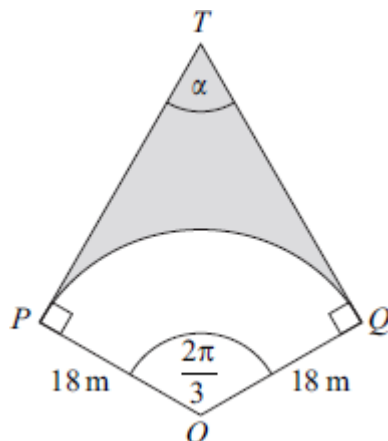


The radius of the circle is 18 m and the angle POQ is $\frac{2\pi}{3}$ radians.

- (a) Find the length of the arc PQ , giving your answer as a multiple of π .

(2)

- (b) The tangents to the circle at the points P and Q meet at the point T , and the angles TPO and TQO are both right angles, as shown in the diagram below.



- (i) Angle $PTQ = \alpha$ radians. Find α in terms of π .

(1)

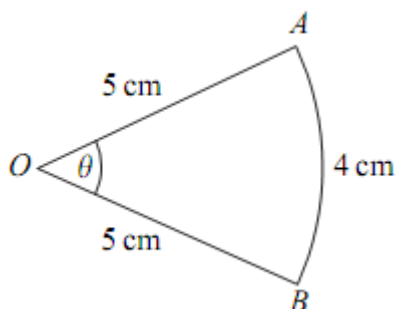
- (ii) Find the area of the shaded region bounded by the arc PQ and the tangents TP and TQ , giving your answer to three significant figures.

(6)

(Total 9 marks)

Q19.

The diagram shows a sector OAB of a circle with centre O and radius 5 cm.



The angle between the radii OA and OB is θ radians.

The length of the arc AB is 4 cm.

- (a) Find the value of θ .

(2)

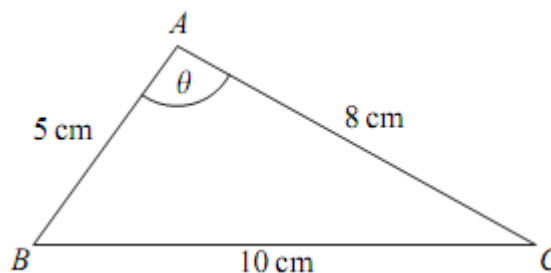
- (b) Find the area of the sector OAB .

(2)

(Total 4 marks)

Q20.

The triangle ABC , shown in the diagram, is such that $AB = 5$ cm, $AC = 8$ cm, $BC = 10$ cm and angle $BAC = \theta$.



- (a) Show that $\theta = 97.9^\circ$, correct to the nearest 0.1° .

(3)

- (b) (i) Calculate the area of triangle ABC , giving your answer, in cm^2 , to three significant figures.

(2)

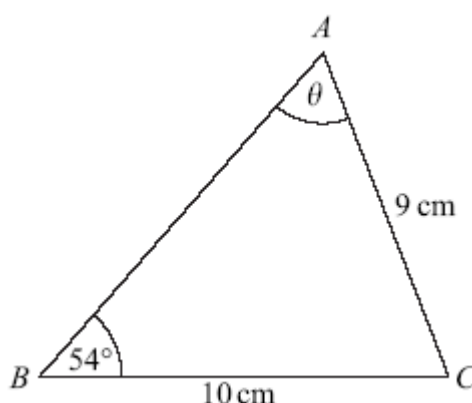
- (ii) The line through A , perpendicular to BC , meets BC at the point D . Calculate the length of AD , giving your answer, in cm, to three significant figures.

(3)

(Total 8 marks)

Q21.

The triangle ABC , shown in the diagram, is such that $AC = 9$ cm, $BC = 10$ cm, angle $ABC = 54^\circ$ and the acute angle $BAC = \theta$.



- (a) Show that $\theta = 64^\circ$, correct to the nearest degree.

(3)

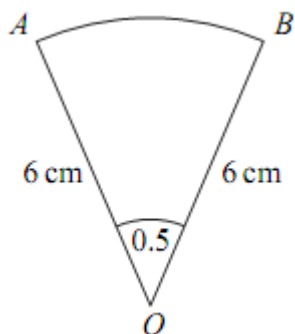
- (b) Calculate the area of triangle ABC , giving your answer to the nearest square centimetre.

(3)

(Total 6 marks)

Q22.

The diagram shows a sector OAB of a circle with centre O .



The radius of the circle is 6 cm and the angle $AOB = 0.5$ radians.

- (a) Find the area of the sector OAB .

(2)

- (b) (i) Find the length of the arc AB .

(2)

- (ii) Hence show that

the perimeter of the sector $OAB = k \times$ the length of the arc AB

where k is an integer.

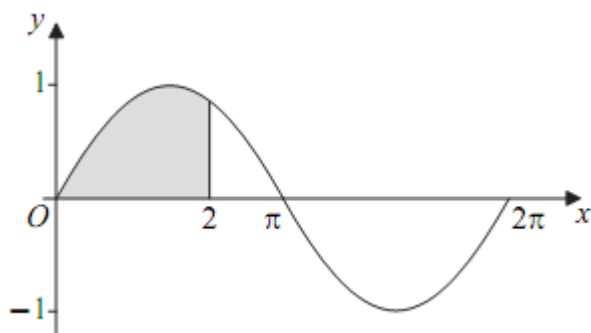
(2)

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(Total 6 marks)

Q23.

A curve C , defined for $0 \leq x \leq 2\pi$ by the equation $y = \sin x$, where x is in radians, is sketched below. The region bounded by the curve C , the x -axis from 0 to 2 and the line $x = 2$ is shaded.



- (a) The area of the shaded region is given by $\int_0^2 \sin x \, dx$, where x is in radians.

Use the trapezium rule with five ordinates (four strips) to find an approximate value for the area of the shaded region, giving your answer to three significant figures.

(4)

- (b) Describe the geometrical transformation that maps the graph of $y = \sin x$ onto the graph of $y = 2 \sin x$.

(2)

- (c) Use a trigonometrical identity to solve the equation

$$2 \sin x = \cos x$$

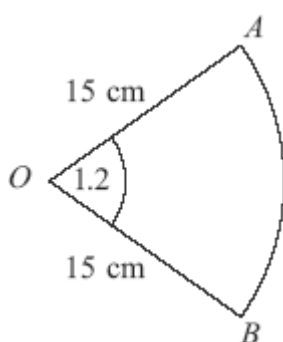
in the interval $0 \leq x \leq 2\pi$, giving your solutions in radians to three significant figures.

(4)

(Total 10 marks)

Q24.

The diagram shows a sector OAB of a circle with centre O .



The radius of the circle is 15 cm and angle $AOB = 1.2$ radians.

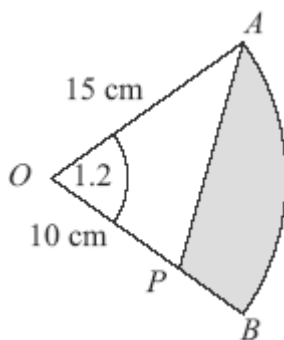
- (a) (i) Show that the area of the sector is 135 cm^2 .

(2)

- (ii) Calculate the length of the arc AB .

(2)

- (b) The point P lies on the radius OB such that $OP = 10$ cm, as shown in the diagram below.



Calculate the perimeter of the shaded region bounded by AP , PB and the arc AB , giving your answer to three significant figures.

(5)

(Total 9 marks)

Q25.

- (a) Solve the equation

$$\operatorname{cosec} x = 3$$

giving all values of x in radians to two decimal places, in the interval $0 \leq x \leq 2\pi$.

(2)

- (b) By using a suitable trigonometric identity, solve the equation

$$\cot^2 x = 11 - \operatorname{cosec} x$$

giving all values of x in radians to two decimal places, in the interval $0 \leq x \leq 2\pi$.

(6)

(Total 8 marks)

Q26.

- (a) Express $\cos x + 3 \sin x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give your value of α , in radians, to three decimal places.

(3)

- (b) (i) Hence write down the minimum value of $\cos x + 3 \sin x$.

(1)

- (ii) Find the value of x in the interval $0 \leq x \leq 2\pi$ at which this minimum occurs, giving your answer, in radians, to three decimal places.

(2)

- (c) Solve the equation $\cos x + 3 \sin x = 2$ in the interval $0 \leq x \leq 2\pi$, giving all solutions,

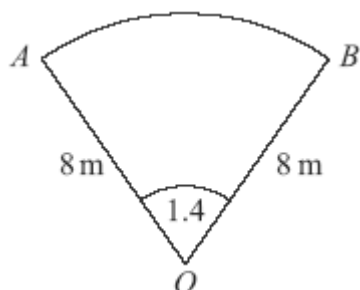
in radians, to three decimal places.

(4)

(Total 10 marks)

Q27.

The diagram shows a sector OAB of a circle with centre O .



The radius of the circle is 8 m and the angle AOB is 1.4 radians.

- (a) Find the area of the sector OAB .

(2)

- (b) (i) Find the perimeter of the sector OAB .

(3)

- (ii) The perimeter of the sector OAB is equal to the circumference of a circle of radius x m. Calculate the value of x to three significant figures.

(2)

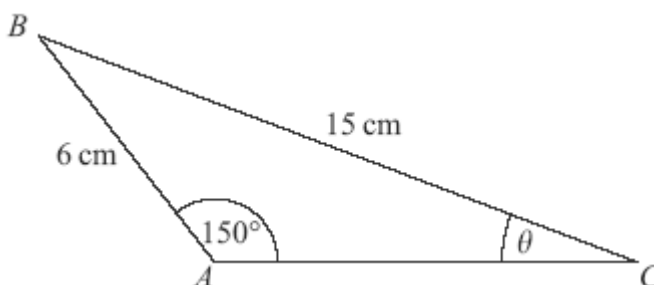
(Total 7 marks)

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Q28.

The triangle ABC , shown in the diagram, is such that $AB = 6$ cm, $BC = 15$ cm, angle $BAC = 150^\circ$ and angle $ACB = \theta$.



- (a) Show that $\theta = 11.5^\circ$, correct to the nearest 0.1° .

(3)

- (b) Calculate the area of triangle ABC , giving your answer in cm^2 to three significant figures.

(3)

(Total 6 marks)

Q29.

- (a) Sketch the graph of $y = \cos x$ in the interval $0 \leq x \leq 2\pi$. State the values of the intercepts with the coordinate axes.

(2)

- (b) (i) Given that

$$\sin^2 \theta = \cos \theta (2 - \cos \theta)$$

prove that $\cos \theta = \frac{1}{2}$.

(2)

- (ii) Hence solve the equation

$$\sin^2 2x = \cos 2x (2 - \cos 2x)$$

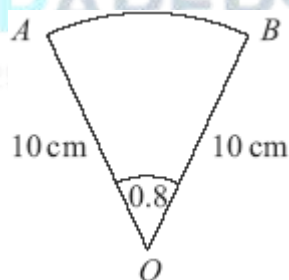
in the interval $0 \leq x \leq \pi$, giving your answers in radians to three significant figures.

(4)

(Total 8 marks)

Q30.

The diagram shows a sector OAB of a circle with centre O and radius 10 cm.



The angle AOB is 0.8 radians.

- (a) Find the area of the sector.

(2)

- (b) (i) Find the perimeter of the sector OAB .

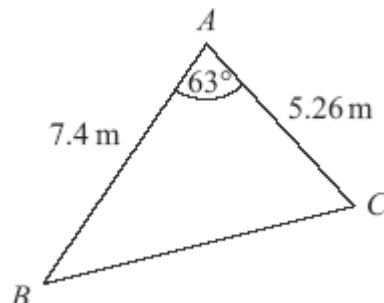
(3)

- (ii) The perimeter of the sector OAB is equal to the perimeter of a square. Find the area of the square.

(2)
(Total 7 marks)

Q31.

The diagram shows a triangle ABC .



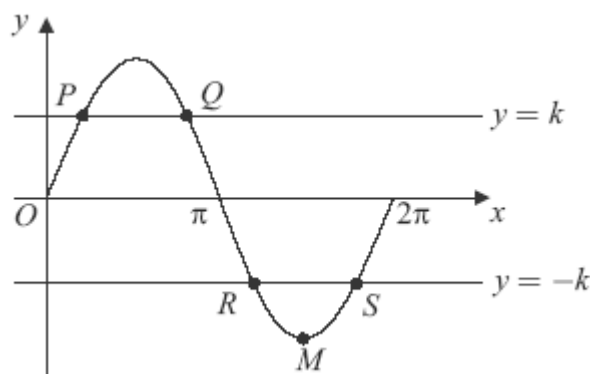
The size of angle A is 63° , and the lengths of AB and AC are 7.4 m and 5.26 m respectively.

- (a) Calculate the area of triangle ABC , giving your answer in m^2 to three significant figures. (2)
- (b) Show that the length of BC is 6.86 m, correct to three significant figures. (3)
- (c) Find the value of $\sin B$ to two significant figures. (2)

(Total 7 marks)

Q32.

- (a) Solve the equation $\sin x = 0.8$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures. (3)
- (b) The diagram shows the graph of the curve $y = \sin x$, $0 \leq x \leq 2\pi$ and the lines $y = k$ and $y = -k$.



The line $y = k$ intersects the curve at the points P and Q , and the line $y = -k$ intersects the curve at the points R and S .

The point M is the minimum point of the curve.

- (i) Write down the coordinates of the point M .

(2)

- (ii) The x -coordinate of P is α .

Write down the x -coordinate of the point Q in terms of π and α .

(1)

- (iii) Find the length of RS in terms of π and α , giving your answer in its simplest form.

(2)

- (c) Sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$, indicating the coordinates of points where the graph intersects the x -axis and the coordinates of any maximum points.

(5)

(Total 13 marks)

Q33.

- (a) Express $\sin x - 3 \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
Give your value of α in radians to two decimal places.

(3)

- (b) Hence:

- (i) write down the minimum value of $\sin x - 3 \cos x$;

(1)

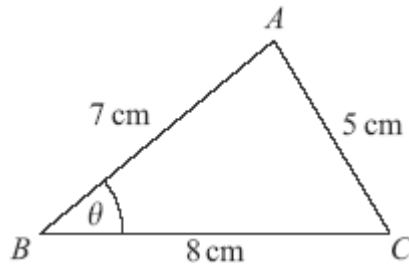
- (ii) find the value of x in the interval $0 < x < 2\pi$ at which this minimum value occurs, giving your value of x in radians to two decimal places.

(2)

(Total 6 marks)

Q34.

The triangle ABC , shown in the diagram, is such that $AB = 7$ cm, $AC = 5$ cm, $BC = 8$ cm and angle $ABC = \theta$.



- (a) Show that $\theta = 38.2^\circ$, correct to the nearest 0.1° .

(3)

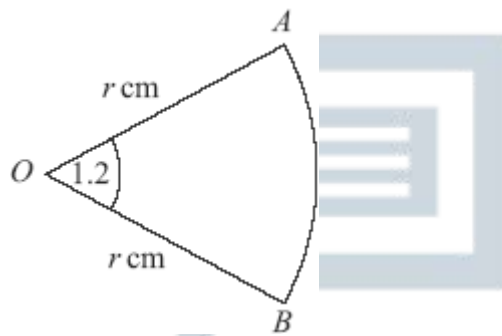
- (b) Calculate the area of triangle ABC , giving your answer, in cm^2 , to three significant figures.

(2)

(Total 5 marks)

Q35.

The diagram shows a sector OAB of a circle with centre O and radius $r\text{ cm}$.



The angle AOB is 1.2 radians. The area of the sector is 33.75 cm^2 .

Find the perimeter of the sector.

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(Total 6 marks)